

Deployment on Cloud Platforms: AWS and Azure

A comprehensive guide for B.Tech Machine Learning students — covering the complete journey from trained ML model to a live, scalable public API on AWS and Azure.

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What is Cloud Deployment & Why Do We Need It?

Simple Definition

Cloud deployment is the process of making an application available on Internet-based cloud infrastructure instead of running it only on a local computer.

Instead of localhost:5000 accessible only to you, your app becomes <https://your-ml-api.com> — accessible to users anywhere.

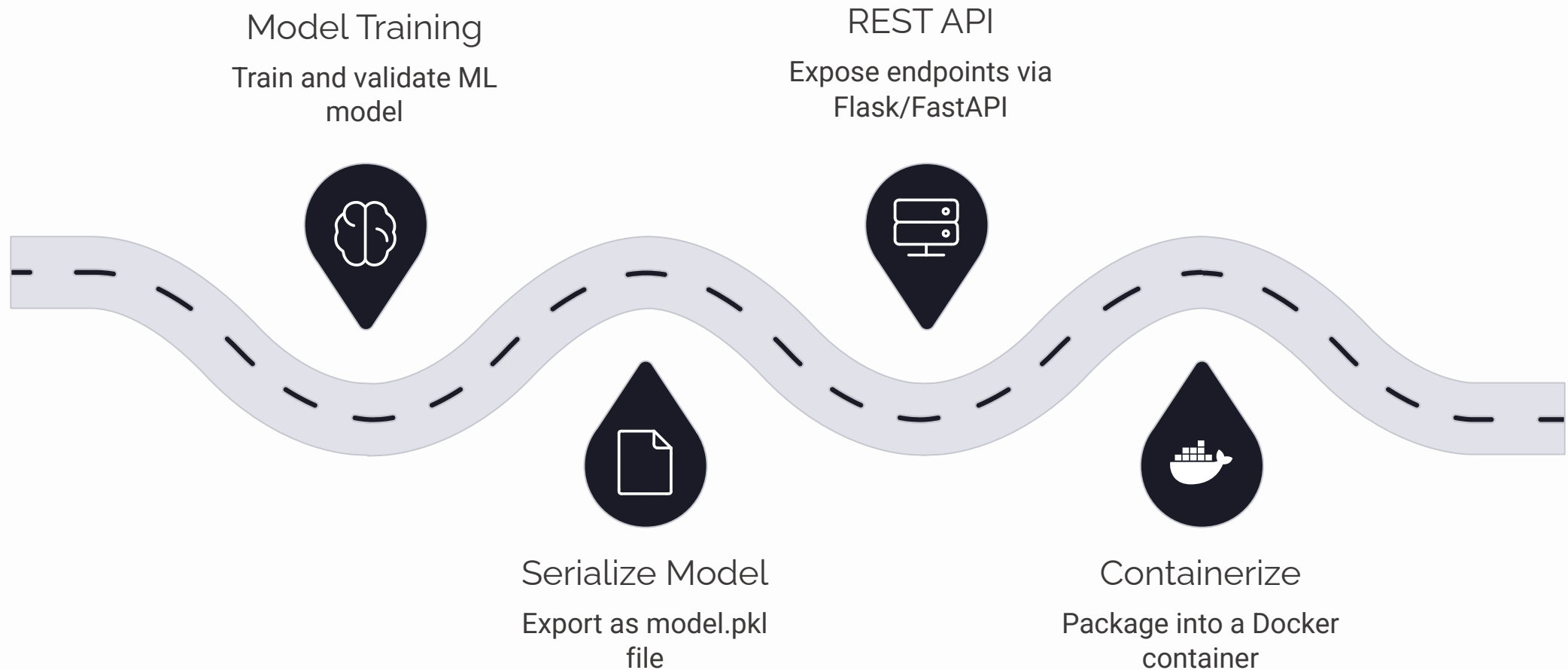
Why Not Just Use a Laptop?

Suppose a student builds a House Price Prediction System. It works locally — but 1,000 users can't use a laptop as a production server. Cloud platforms provide:

- Computing power & storage
- Networking & security
- Scalability & high availability
- Databases & monitoring

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The Complete ML Deployment Journey



This pipeline is the backbone of every production ML system. Each step builds on the previous — model serialization, REST API creation, containerization, and finally cloud deployment work together to deliver predictions to real users at scale.

AWS vs Azure: Two Major Cloud Platforms

Amazon Web Services (AWS)

EC2

Virtual cloud server

S3

Object storage

ECR

Container registry

ECS

Container service

SageMaker

ML platform

Microsoft Azure

Azure VM

Virtual machines

Blob Storage

Object storage

ACR

Container registry

Container Apps

Run containers

Azure ML

ML platform

AWS & Azure: Service-by-Service Comparison

Requirement	AWS	Azure
Virtual server	EC2	Virtual Machines
Object storage	S3	Blob Storage
Container registry	ECR	Azure Container Registry
Container service	ECS	Container Apps
Kubernetes	EKS	AKS
ML platform	SageMaker	Azure Machine Learning
Managed SQL DB	RDS	Azure SQL Database

These aren't identical products, but they solve similar categories of problems. Students can choose either platform – the concepts transfer directly.

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Docker's Role: Packaging Before Cloud

Docker ≠ Competitor to AWS/Azure

Students often think Docker and AWS are competing technologies. They are not — they solve different problems:

Docker

Packages the application and its dependencies into a portable container image.

AWS / Azure

Provides cloud infrastructure and services to run, scale, secure, and monitor that container.

Deploying a Flask ML App on EC2

For a project with `app.py`, `model.pkl`, `requirements.txt`, and a `Dockerfile`:

01

Push code to AWS EC2

02

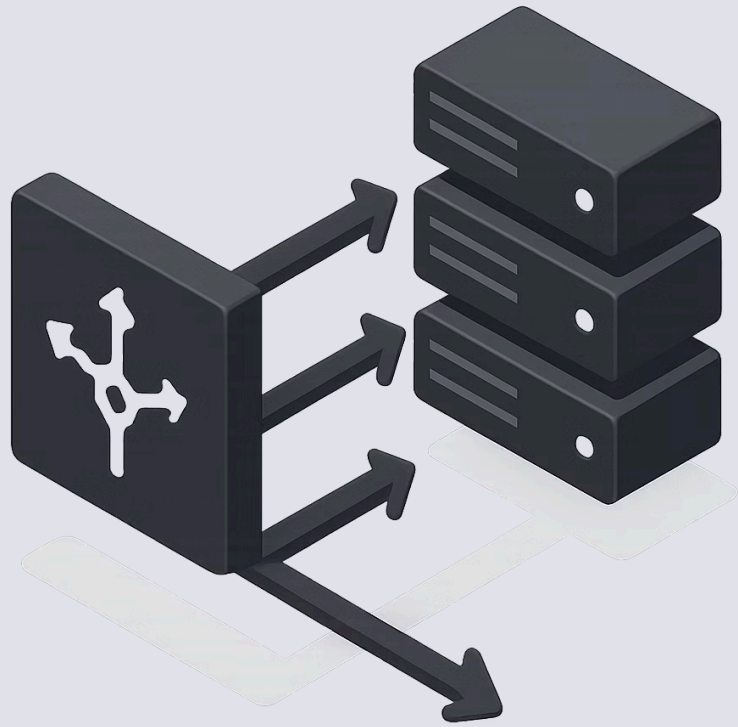
Install Docker on EC2

03

Build / Pull Docker Image

04

Run Container → Flask API live on Internet



Scaling, Security & Monitoring



Scaling

Vertical: Increase CPU/RAM of one server. **Horizontal:** Add more containers behind a Load Balancer. Cloud-native systems prefer horizontal scaling for high availability.



Security

Use **HTTPS** (not HTTP), **Authentication & Authorization**, **Firewall/Security Groups**, and **Environment Variables** for secrets — never hardcode API keys in source code.



Monitoring

Track: Is the app running? Request count, CPU/memory usage, error rates, response times. Cloud platforms provide built-in logging and monitoring dashboards.

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CI/CD & Complete Industry Architecture

CI/CD Pipeline

In industry, developers don't manually deploy every change. The automated workflow is:

01

Developer pushes code to GitHub

02

CI/CD pipeline triggers automatically

03

Build Docker Image & run tests

04

Push image → Deploy to AWS/Azure

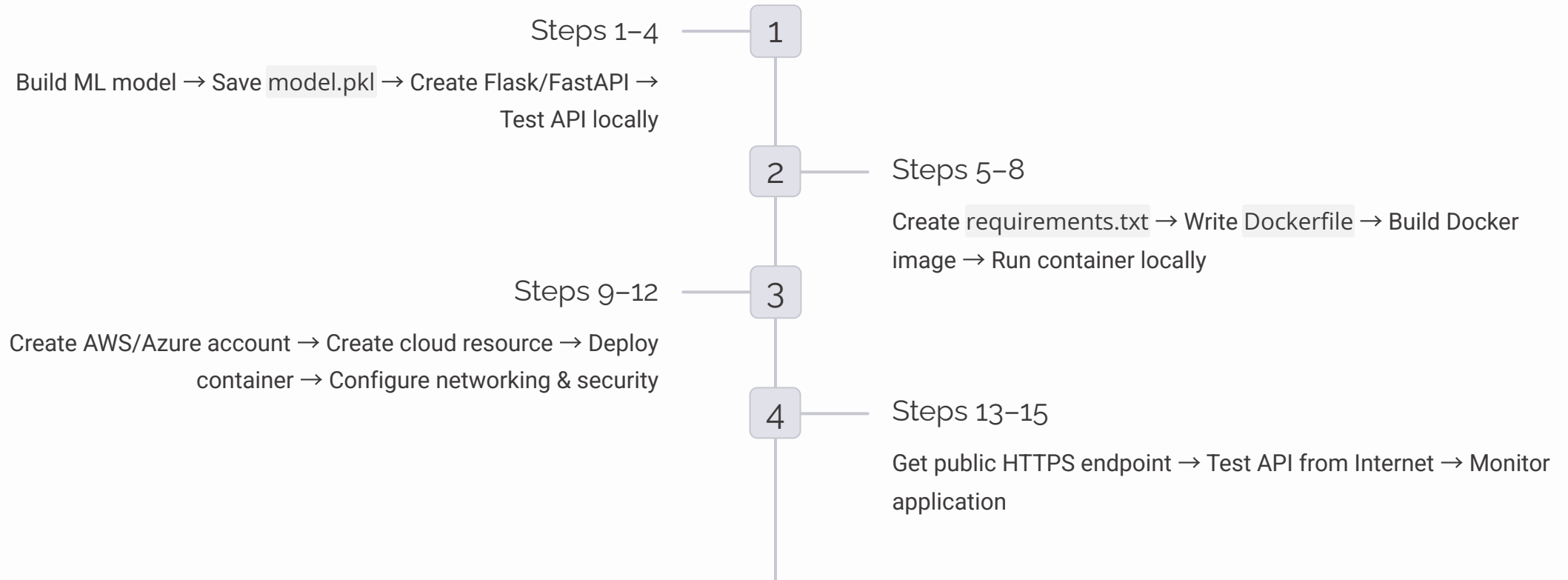
Industry ML Architecture

A production system includes multiple layers working together:

- **Load Balancer** distributes traffic across containers
- **ML API containers** serve predictions
- **Container Registry** stores Docker images
- **Database** stores user data and results
- **Monitoring** tracks logs and metrics
- **Security** – IAM, network rules, secrets management

❏ A production system needs much more than simply making the Flask port publicly accessible.

Student Project Roadmap: 15 Steps to Production



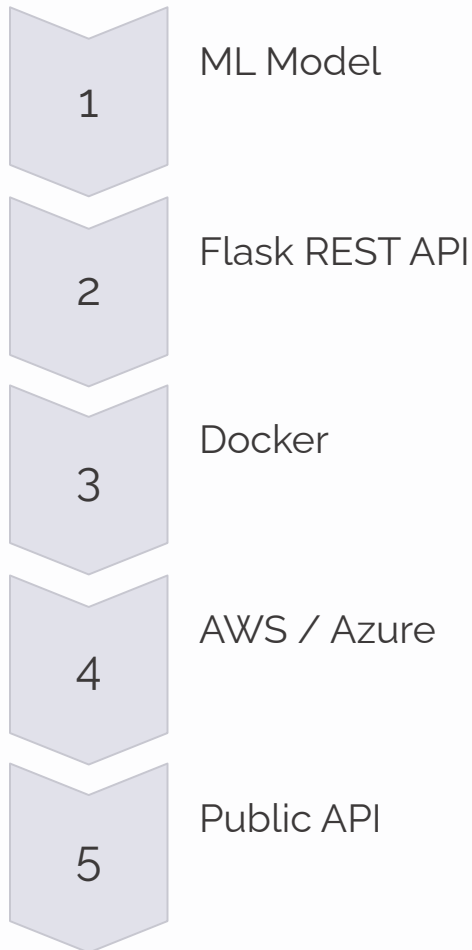
Example project: **Diabetes Prediction ML Application** — `model.pkl` + `app.py` + `Dockerfile` → Docker Image → AWS ECR / Azure ACR → Container Service → Public HTTPS Endpoint → Users. This is the complete journey from ML model to production application.

The Big Picture & Key Interview Q&A

★ Definition to Remember

Cloud deployment is the process of deploying an application on cloud infrastructure such as AWS or Azure so that it can be accessed by users over the Internet. Docker packages the application, while AWS or Azure provides the infrastructure to run, scale, secure, and monitor it.

Best Practical Sequence



Interview Questions

What is cloud deployment?

Making an app available via cloud infrastructure so users can access it over the Internet.

What is AWS EC2 / Azure VM?

Services that provide virtual computing instances in the cloud.

What is ECR / ACR?

Managed registries for storing Docker container images.

What is scaling?

Increasing capacity to handle more workload – vertically (bigger server) or horizontally (more containers).

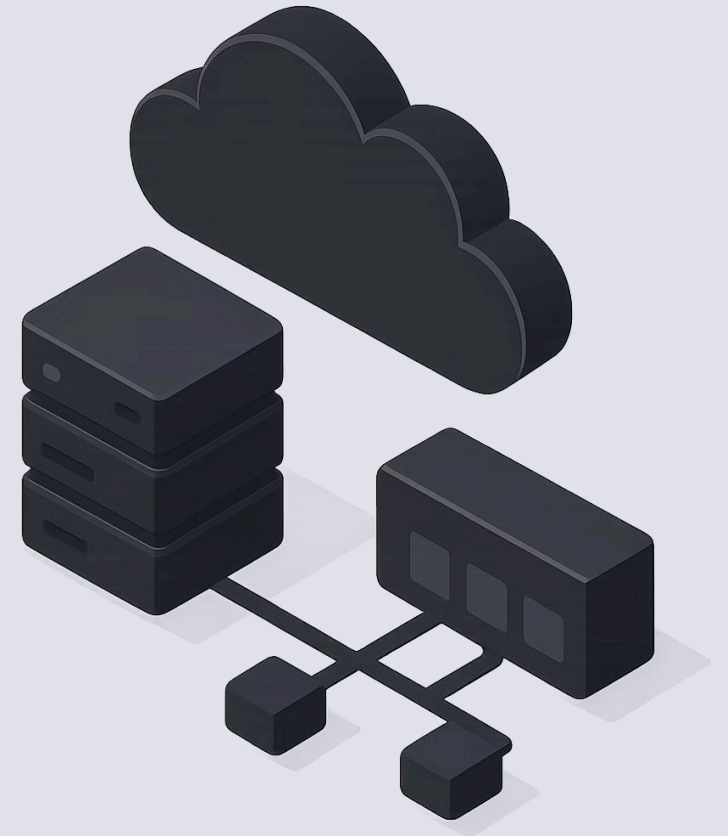
What is CI/CD?

Automated pipeline for building, testing, and deploying software continuously.

AWS and Azure — From Basic to Advanced

A structured guide for B.Tech Machine Learning students — understanding cloud computing from first principles to production ML deployment.

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1. What is Cloud Computing?

The Traditional Way

To build an ML application traditionally, you buy a physical server. This requires: Hardware purchase, Electricity, Networking, Cooling, Maintenance, System administration, Hardware upgrades.

The Cloud Way

Instead of buying the server, you rent computing resources from a cloud provider over the Internet.

Your Laptop → Internet → Cloud Provider → Compute, Storage, Database, Networking, Security, AI/ML Services

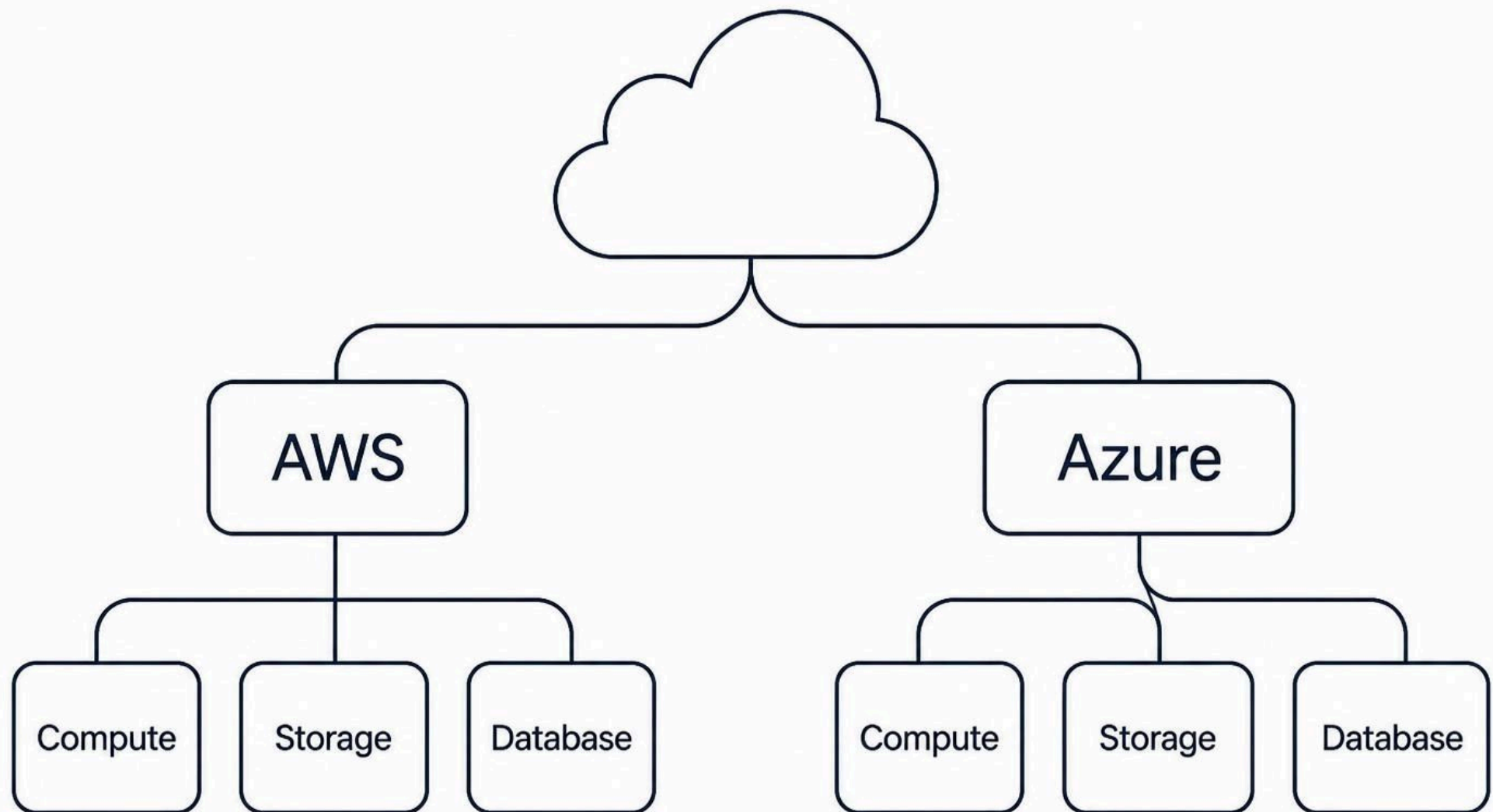
"Cloud computing means using computing resources such as servers, storage, databases and software over the Internet instead of owning and managing all the physical infrastructure yourself."

2. What are AWS and Azure?

Two major cloud platforms:

- **AWS:** Amazon Web Services — provided by Amazon
- **Azure:** Microsoft Azure — provided by Microsoft

Both provide hundreds of cloud services. Think of them as large collections of Internet-accessible computing infrastructure and managed services.



AWS (Amazon Web Services)

Pioneering cloud platform, offering a vast array of services from computing power to databases and machine learning.



Azure (Microsoft Azure)

Microsoft's comprehensive cloud offering, providing solutions for computing, analytics, storage, and networking.

3. A Simple Real-Life Analogy

Option 1: Build Everything Yourself

Like buying land, building, kitchen, electricity equipment, water system, furniture – and maintaining everything yourself.

Option 2: Rent a Ready Facility

Rent a professionally managed facility and concentrate on running the restaurant.

❏ Cloud computing is conceptually similar to renting computing infrastructure/services rather than purchasing and maintaining all the underlying hardware yourself.

4. Why Do Companies Use AWS/Azure?

Suppose a company develops an e-commerce website. It needs: Website, Backend, Database, Images, Videos, Authentication, Security, Monitoring, Backup.

Instead of buying and maintaining dozens of physical servers, it can use cloud services.



Compute

Run programs and applications



Storage

Store files, images, videos, datasets



Databases

Store structured application data



Networking

Connect applications, servers and users



Security

Control access and protect resources



AI/ML

Build, train and deploy AI/ML systems



Monitoring

Monitor applications and infrastructure

5-8. Compute, EC2, Azure VMs & Storage

Compute: EC2 & Azure VM

In cloud computing, you can rent computing resources (CPU, RAM, OS) from the cloud.

- **AWS EC2 (Elastic Compute Cloud):** A virtual server you rent inside AWS. You can run Python, Flask, FastAPI, ML applications, web servers.
- **Azure Virtual Machines:** Azure's equivalent. AWS EC2 $\approx$ Azure Virtual Machine.

Storage: S3 & Azure Blob

Cloud platforms provide object storage for: Images, Videos, PDFs, Datasets, ML Models, Backups, Logs.

- **Amazon S3 (Simple Storage Service):** Organized into S3 Buckets containing files and folders.
- **Azure Blob Storage:** Organized into Storage Accounts $\rightarrow$ Containers $\rightarrow$ Files.

9–13. Databases, Networking & Regions

Databases

Most applications need databases. Cloud providers offer managed database services.

- **AWS:** Amazon RDS, Amazon DynamoDB
- **Azure:** Azure SQL Database, Azure Cosmos DB

Why "Managed"? The cloud handles installation, updates, backups, security, maintenance, scaling, monitoring. You focus on schema, data, queries and application logic.

Networking: VPC & VNet

Components (Web, Backend, ML API, Database) must communicate. Cloud platforms provide networking.

- **AWS VPC (Virtual Private Cloud):** Isolated virtual network for AWS resources. Web Server | API Server | Database all inside VPC.
- **Azure VNet (Virtual Network):** Azure's equivalent. AWS VPC $\approx$ Azure VNet.

Regions & Availability Zones

Cloud providers have data centers in different geographic locations.

- **Region:** A geographic area containing cloud infrastructure. Choose a region close to your users.
- **Availability Zone (AZ):** Isolated locations within a region. Multiple AZs provide resilience — if one AZ has a problem, the application can continue operating.

14. AWS and Azure: Service Categories

Learn services by category, rather than trying to memorize hundreds of names.

Category	AWS	Azure
Virtual server	EC2	Virtual Machines
Object storage	S3	Blob Storage
Relational DB	RDS	Azure SQL Database
Virtual network	VPC	VNet
Container registry	ECR	Azure Container Registry
Kubernetes	EKS	AKS
ML platform	SageMaker	Azure Machine Learning
Serverless functions	Lambda	Azure Functions
Monitoring	CloudWatch	Azure Monitor
Identity	IAM	Microsoft Entra ID / Azure RBAC

15–16. Serverless, Containers & Cloud + Docker

Serverless Computing

Suppose you have a small function. You don't necessarily need to maintain a complete server yourself.

- **AWS Lambda:** Serverless compute on AWS
- **Azure Functions:** Serverless compute on Azure

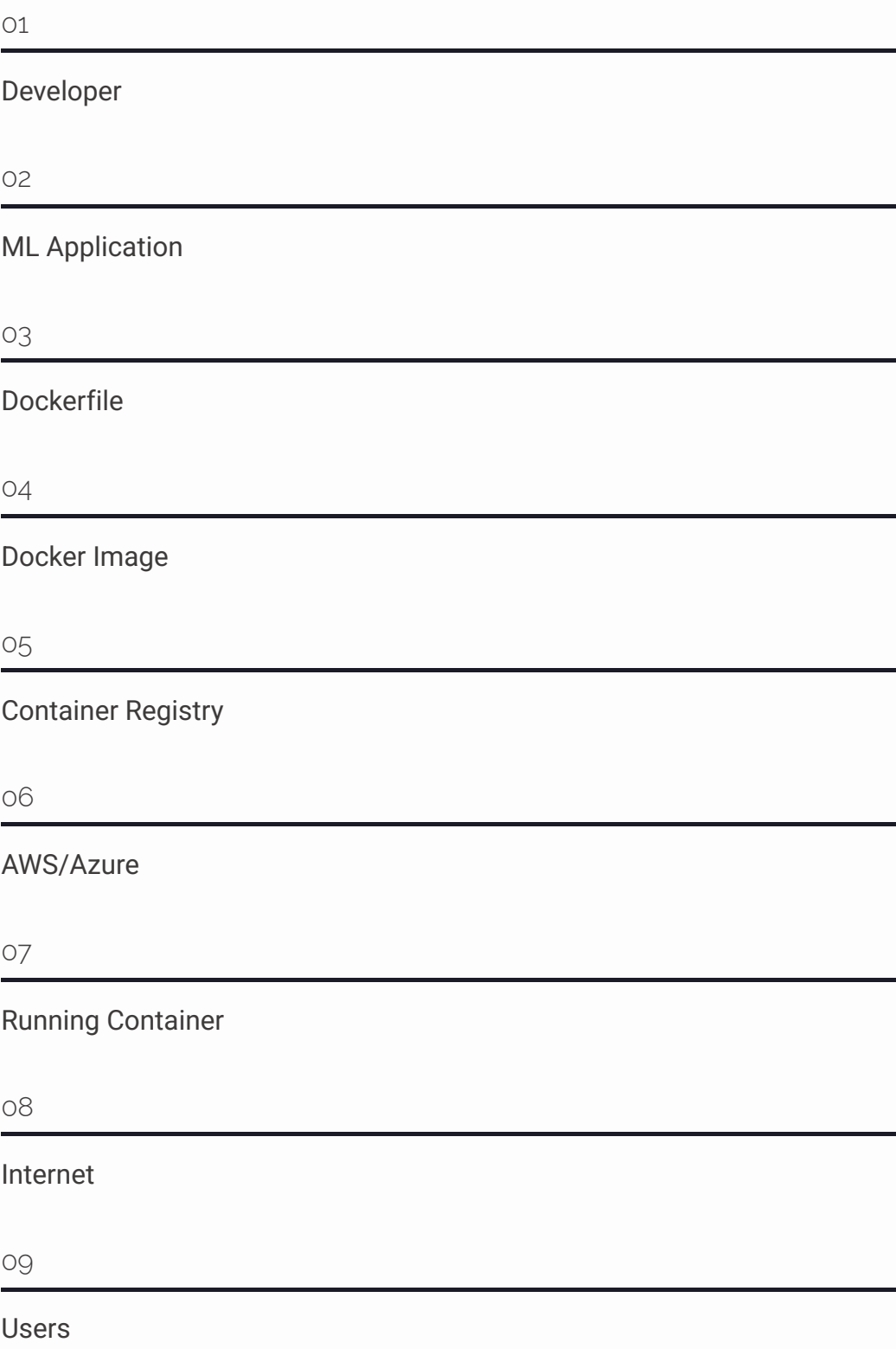
You focus mainly on the code while the platform manages much of the underlying server infrastructure.



Cloud + Docker: The Connection

A container packages: Application + Dependencies + Runtime (Python, Flask, NumPy, Pandas, ML Model, Application).

The complete flow:



AWS path: Docker Image → Amazon ECR → ECS / EKS / other compute

Azure path: Docker Image → Azure Container Registry → Container Apps / AKS / other compute

17–18. Container Registry & Kubernetes

Container Registry

A container registry stores Docker/container images.

- **AWS:** Amazon ECR (Elastic Container Registry)
- **Azure:** Azure Container Registry (ACR)

Think of it as: "A cloud warehouse for container images."

Flow: Developer → docker build → Docker Image → Registry → Cloud Service → Container

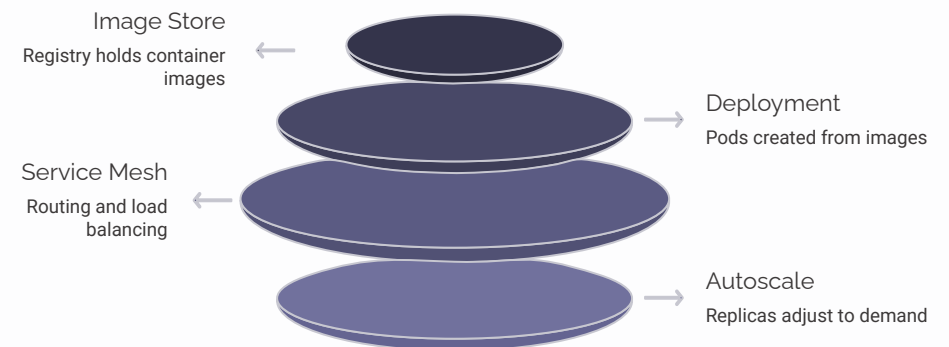
Kubernetes

When an organization has hundreds or thousands of containers, manually managing them becomes difficult. Kubernetes helps orchestrate containers.

- **AWS:** Amazon EKS (Elastic Kubernetes Service)
- **Azure:** Azure Kubernetes Service (AKS)

Kubernetes can help with:

- Scaling
- Scheduling
- Service discovery
- Rolling deployments
- Self-healing
- Container management

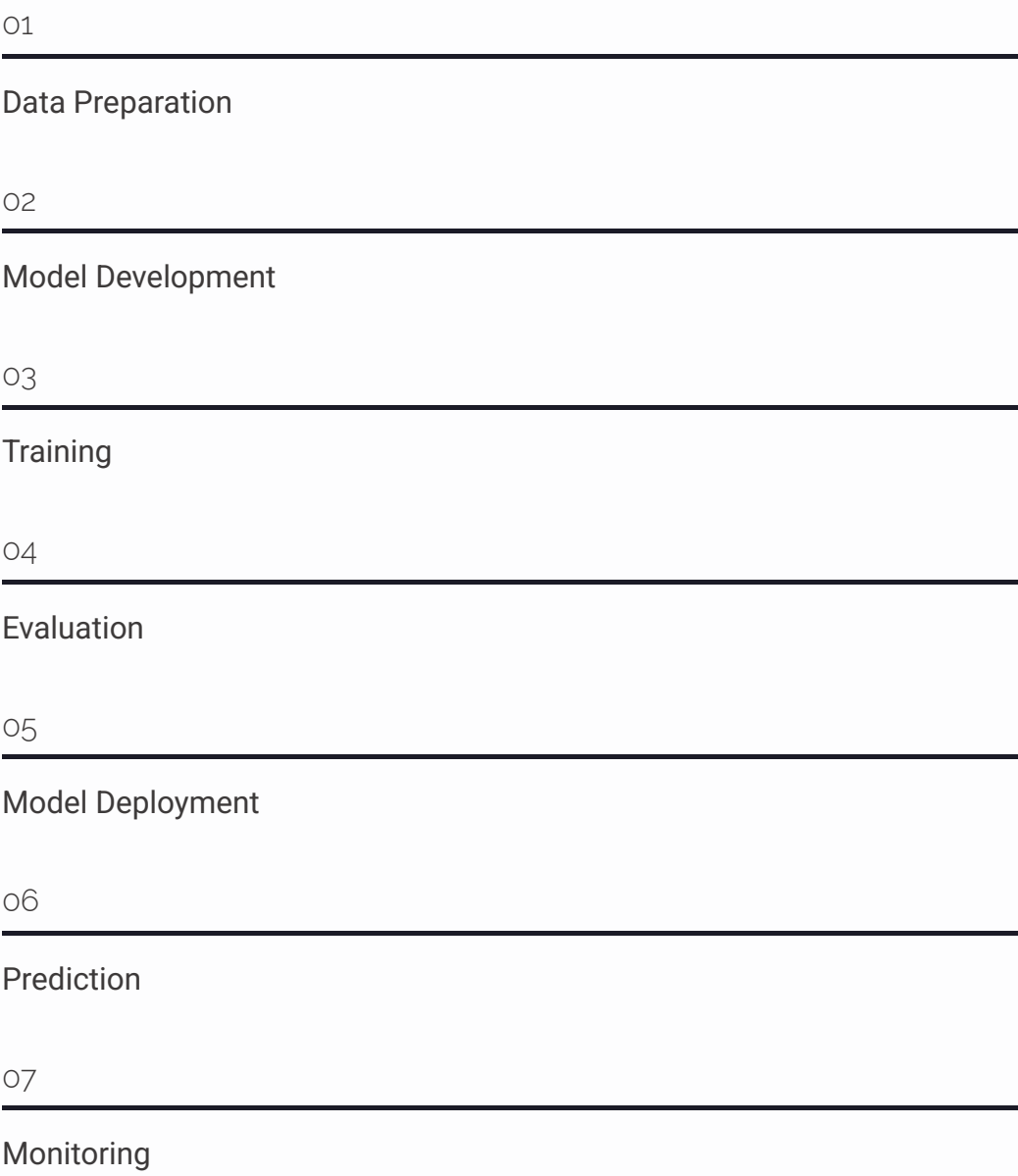


19–20. ML Platforms: SageMaker & Azure ML

AWS SageMaker

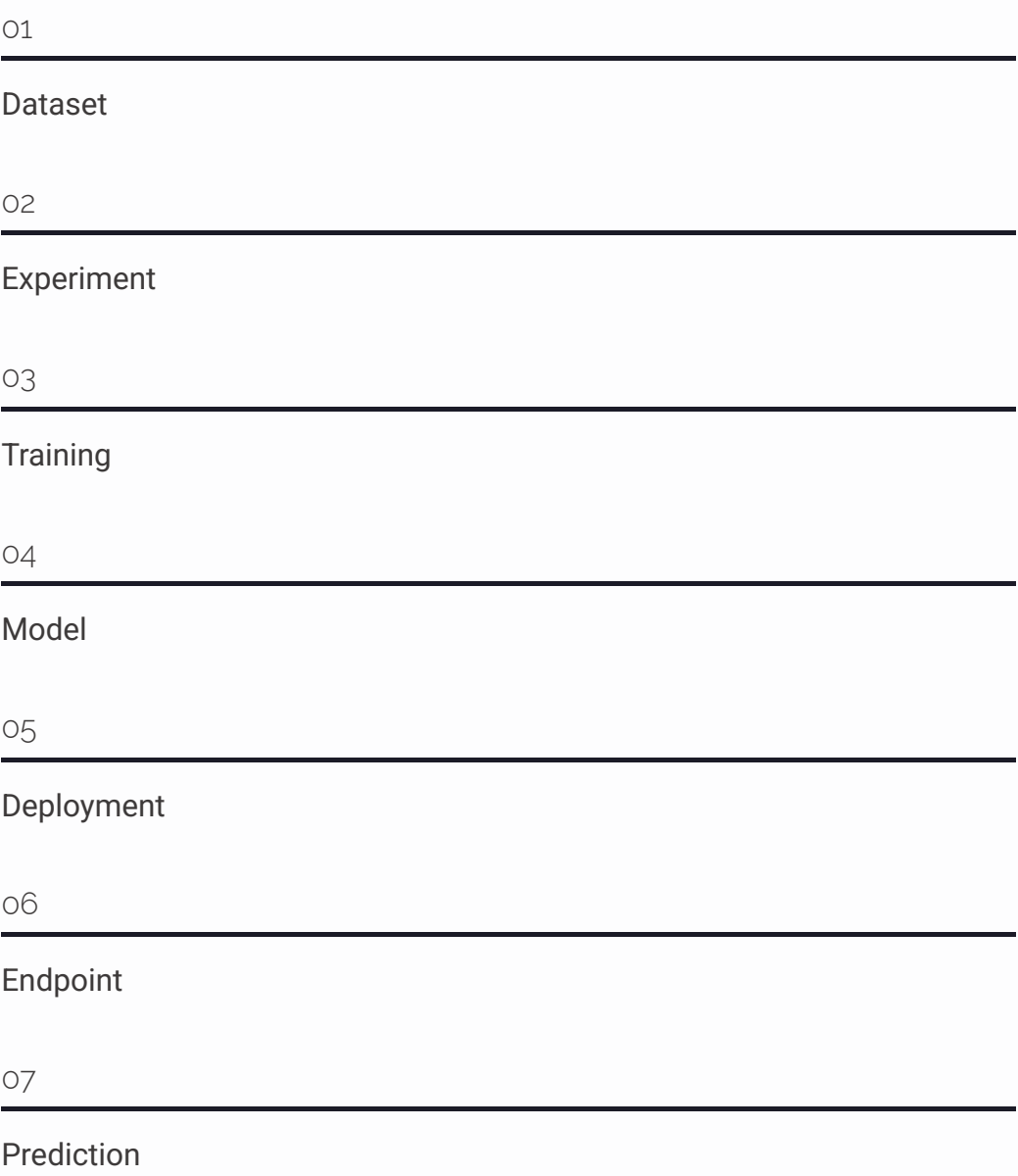
Amazon SageMaker is an AWS platform/service family designed specifically to help build, train, deploy and operate machine-learning models.

Workflow:



Azure Machine Learning

Azure Machine Learning supports ML workflows:



 **AWS SageMaker ≈ Azure Machine Learning** Both platforms support cloud-based ML workflows: ML Development, Training, Deployment, Monitoring.

21–22. AI Services & Generative AI on Cloud

Ready-Made AI Services

Cloud platforms provide ready-made AI services. Organizations may use cloud APIs/services for:

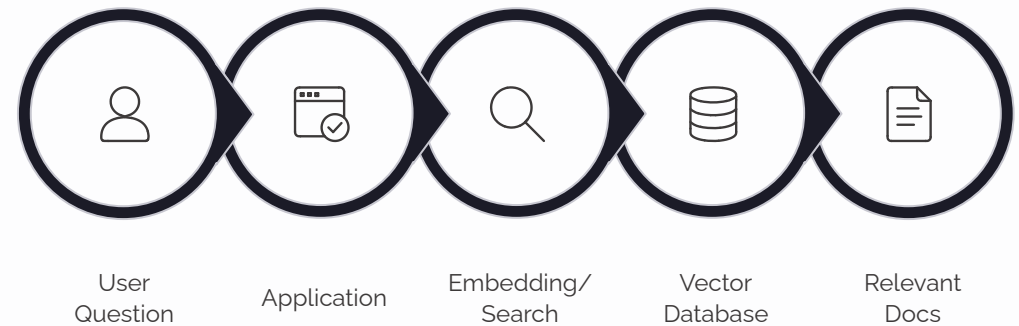
- Computer vision
- Speech
- Natural language processing
- Generative AI
- Document processing
- Search
- Recommendation systems

This means you don't always have to train a model from scratch.

Generative AI Architecture

Modern cloud platforms provide services for developing generative-AI applications.

A typical RAG application architecture:



Cloud infrastructure can host: the application, databases, model endpoints and supporting services.

23–25. IAM, Security Groups & Load Balancing

Identity & Access Management

Not all users should have the same permissions.

- **AWS:** IAM (Identity and Access Management)
- **Azure:** Microsoft Entra ID + Azure RBAC

Concept: User → Authentication → Authorization → Allowed Resources

Example roles: Admin (full access), Developer (can deploy), Student (can use app), Guest (limited access)

Security Groups & Firewalls

Not all ports should be exposed to the Internet. Network security rules determine what traffic is allowed.

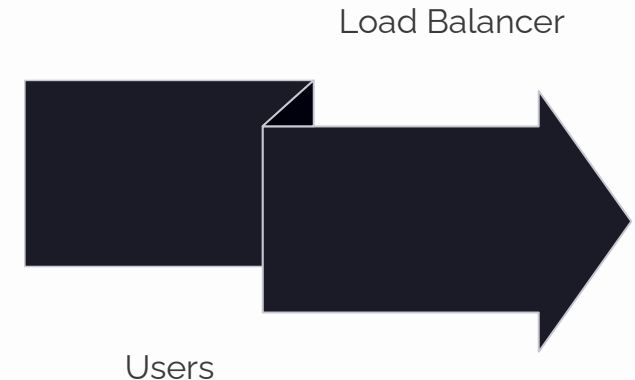
Internet → Firewall / Security Rules → Allowed | Blocked

For production: HTTPS and carefully controlled network access are essential.

Never hardcode API keys in source code — use Environment Variables for secrets.

Load Balancer & Auto Scaling

Load Balancer: Distributes incoming requests across available application instances.



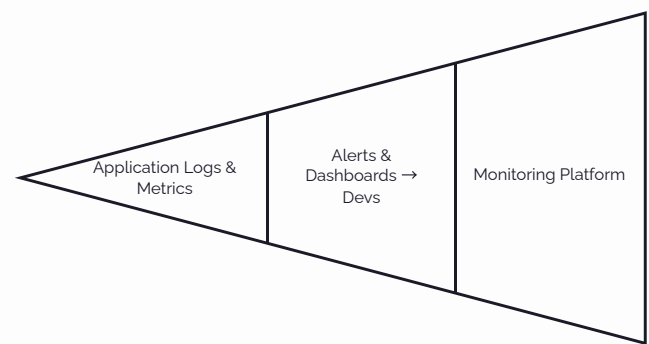
26–28. Monitoring, Backup & Cloud Cost Model

Monitoring

After deployment, you need to monitor the system. Key questions: Is the server running? CPU usage? Memory? Request count? Error rates? Response time? Is the ML API failing?

- **AWS:** CloudWatch
- **Azure:** Azure Monitor

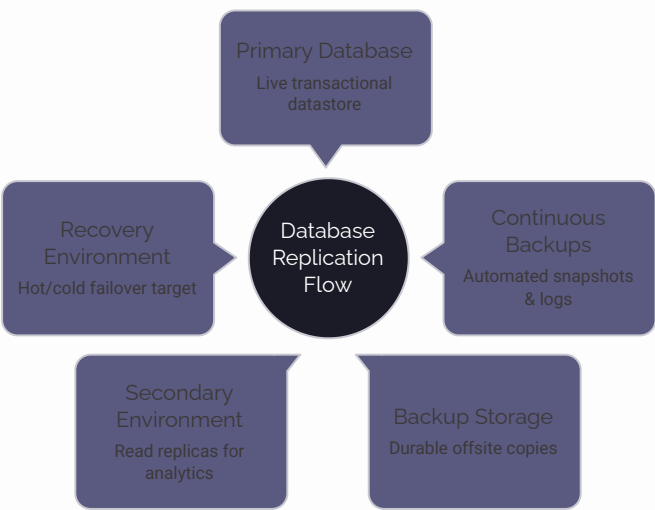
Flow: Application → Logs + Metrics → Monitoring Platform → Alerts / Dashboard → Developer / DevOps



Backup & Disaster Recovery

Suppose a database server fails. What happens to your data? Cloud systems can provide: Backups, Snapshots, Replication, Multi-zone deployment, Disaster recovery.

Flow: Primary Database → Backup → Secondary / Recovery Environment



The exact design depends on business requirements.

Cloud Cost Model

Traditional: Buy Server → Pay Large Upfront Cost

Cloud: Use Resources → Pay According to Service/Usage/Plan

Important warning: Cloud is NOT automatically cheap. Poorly configured resources can generate significant costs. Example: Unused VM running 24×7 = 💰 wasted money.

Cloud cost optimization is an important industry skill.

29–30. Infrastructure as Code & DevOps + AWS/Azure

Infrastructure as Code (IaC)

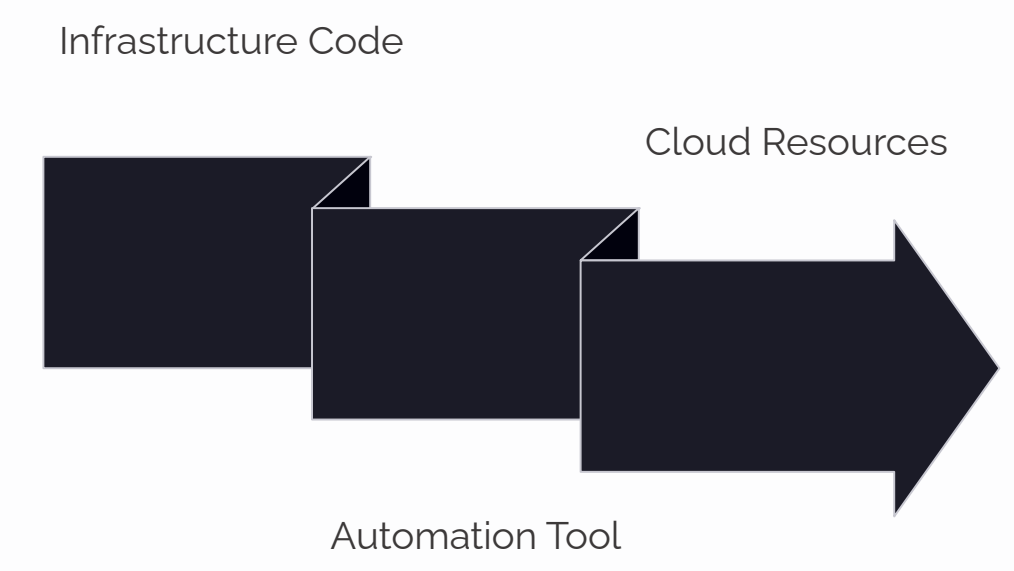
Instead of manually creating resources through a web portal, infrastructure can be described as code.

You can define: Network, Server, Database, Security Rules, Load Balancer – all as code.

Popular technologies:

- Terraform
- AWS CloudFormation
- Azure Bicep

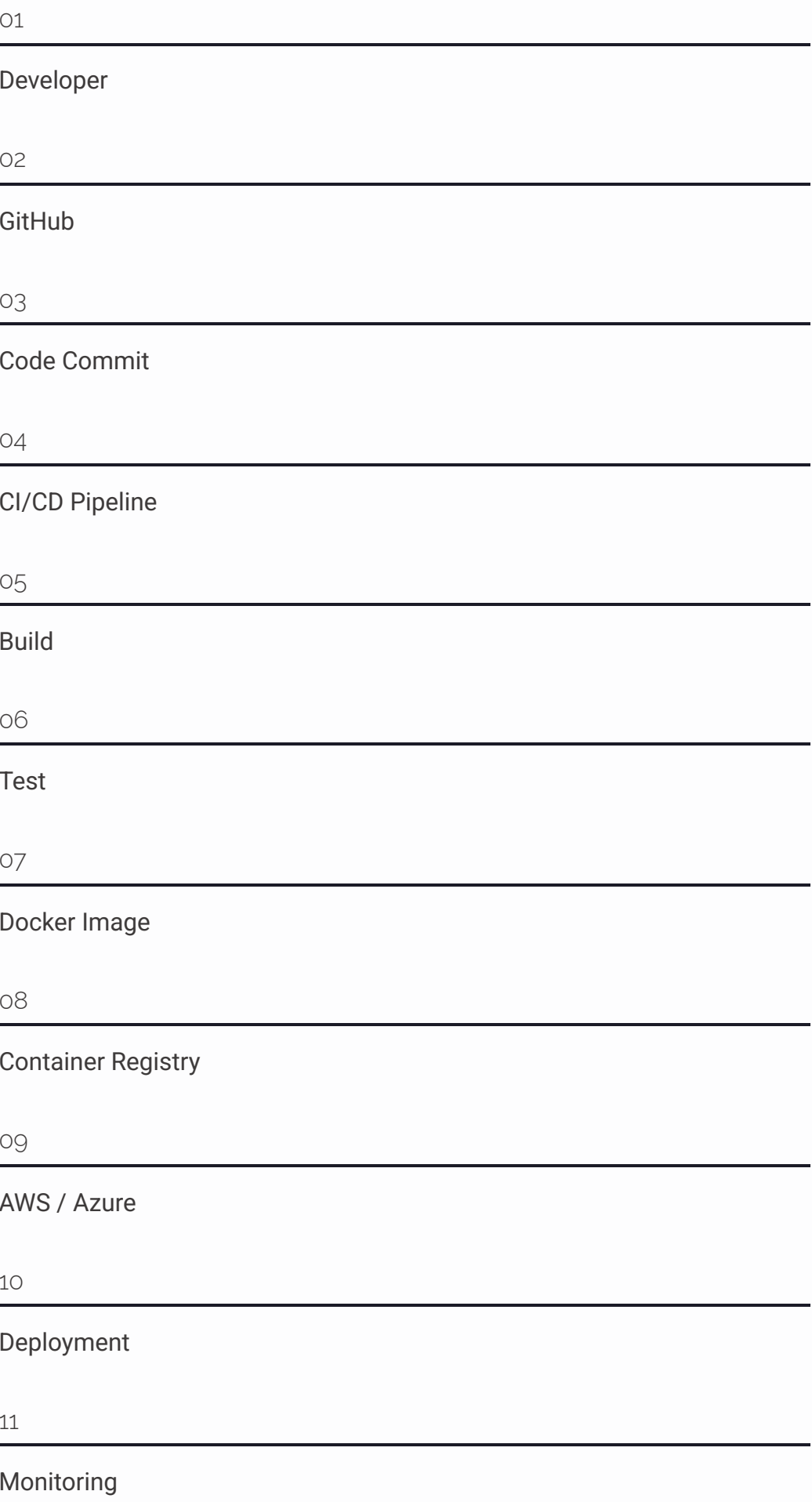
Flow:



This makes infrastructure reproducible and easier to manage.

DevOps + AWS/Azure Pipeline

In industry, cloud is strongly connected with DevOps. A typical CI/CD pipeline:

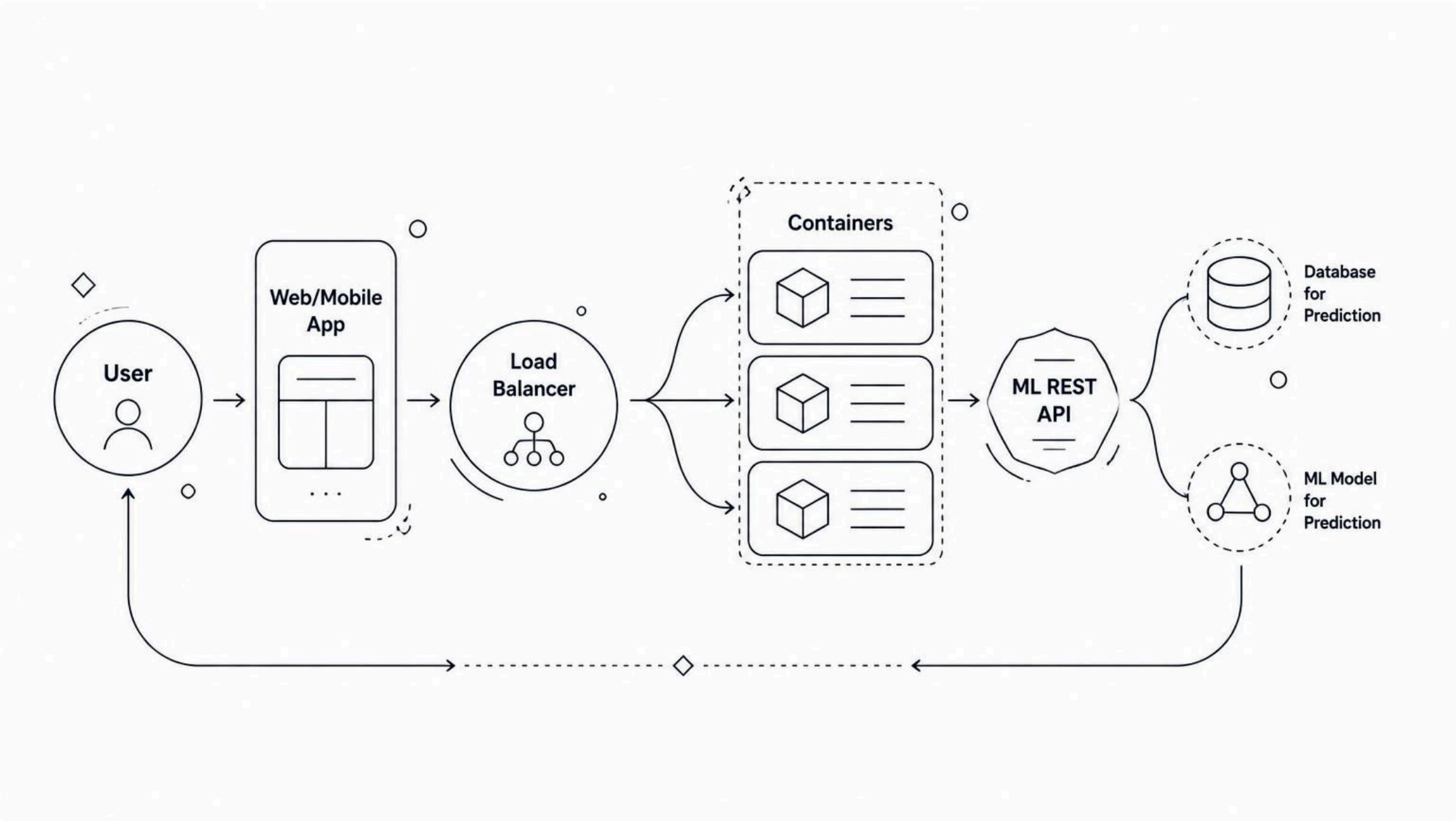


This is much closer to how professional software systems are operated.

31. Complete ML Industry Architecture

The complete cloud-native view of an ML application:

USER → Web / Mobile App → HTTPS/API → Load Balancer → [Container | Container | Container] → ML REST API → [ML Model | Database] → Prediction → User



Supporting Services

 Container Registry
Stores Docker Images

 Identity & IAM
Ensures Security

 CI/CD
Automates Deployment

 Object Storage
Holds Datasets/Models

 Monitoring
Collects Logs + Metrics

 Kubernetes/Container Service
Manages Scaling

32. AWS vs Azure — Simple Mental Model

Don't memorize hundreds of services. Learn this mapping:

Requirement	AWS	Azure
Compute	EC2	VM
Storage	S3	Blob
Database	RDS	Azure SQL
Container Registry	ECR	ACR
Container Orchestration	ECS/EKS	Container Apps/AKS
Machine Learning	SageMaker	Azure ML

33. Basic → Intermediate → Advanced Learning Path

Level 1 — Cloud Basics

Cloud Computing → AWS, Azure → Regions, Availability Zones

Level 2 — Core Services

Compute, Storage, Database, Networking

Learn: EC2 / Azure VM | S3 / Blob | RDS / Azure SQL | VPC / VNet

Level 3 — Application Deployment

Python → Flask/FastAPI → Docker → Container Registry → Cloud

Level 4 — Production

Load Balancer, Scaling, Monitoring, Security, HTTPS, IAM

Level 5 — Advanced

Kubernetes, CI/CD, Infrastructure as Code, Microservices, Serverless, Cloud-native architecture

Level 6 — AI/ML

ML Training → Model Registry → Model Deployment → Inference Endpoint → Monitoring → Retraining

34. What Should B.Tech Students Actually Learn?

Students do not need to become AWS/Azure administrators to complete a good ML project.

01	02	03
Cloud Computing	AWS / Azure basics	Virtual Machine
04	05	06
Storage	Database	Docker
07	08	09
Container Registry	Flask/FastAPI	Deploy ML API
10	11	
Security	Monitoring	

📌 For an ML project, understanding these 11 concepts gives students a solid foundation for real-world cloud deployment.

35. Example Student Project: AI-Based Crop Disease Detection

Show the complete project journey in a visual flow:

Project: AI-Based Crop Disease Detection



Stage 1 — Development

Python → PyTorch/TensorFlow → model.pth

Input: Image → ML/DL Model → Prediction



Stage 2 — API

FastAPI → POST /predict endpoint



Stage 3 — Docker

FastAPI + Model + Dependencies → Docker Image



Stage 4 — Cloud

Docker Image → AWS / Azure → Cloud Container → Public API →
Mobile/Web App

Key transformation callout:



Before: "My model runs on my laptop."

After: "My ML model is available as an Internet-accessible service."

"That is the key idea of cloud deployment."

36. Final Summary & The Simplest Explanation

Final Concept Map

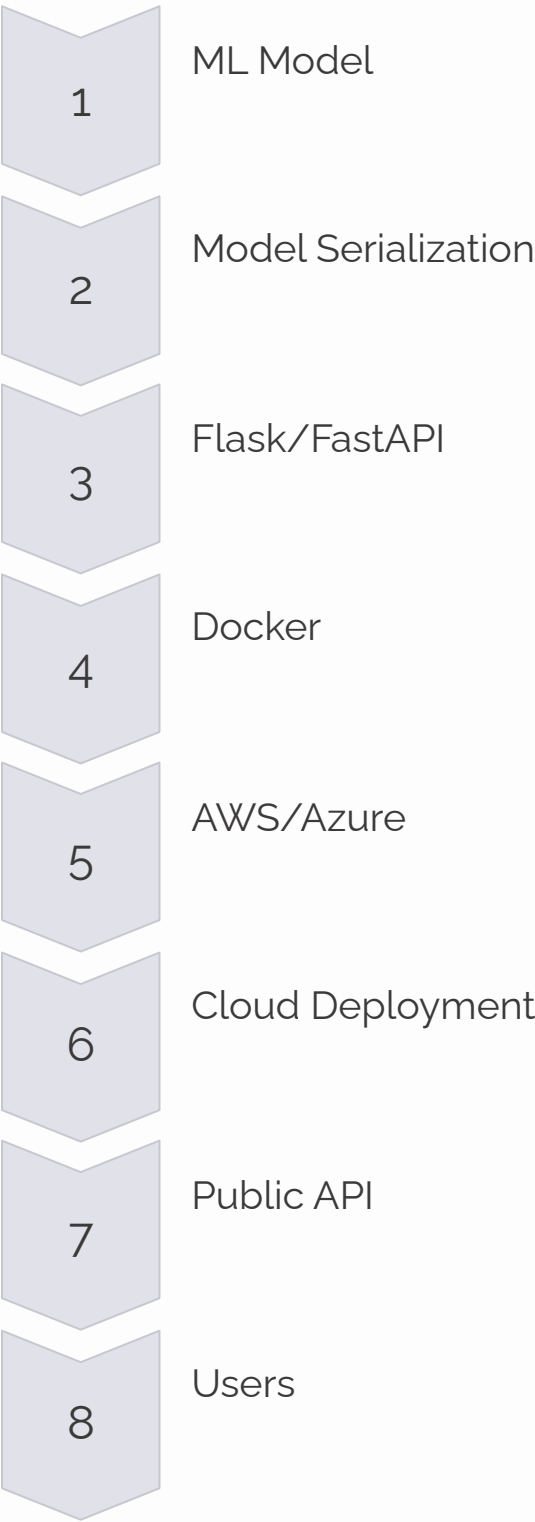
CLOUD COMPUTING → AWS | Azure

- **AWS:** Compute (EC2) | Storage (S3) | Database (RDS) → Containers → ECR → ECS/EKS → ML Application
- **Azure:** Compute (VM) | Storage (Blob) | Database (Azure SQL) → Containers → ACR → Container Apps/AKS → ML Application

★ The Simplest Explanation for Students

AWS and Azure are cloud computing platforms that provide computing power, storage, databases, networking, security, AI/ML services, containers and many other IT resources over the Internet. AWS is provided by Amazon, while Azure is provided by Microsoft.

And the complete ML connection:



That sequence gives students a very clear understanding of how an ML model moves from a Python notebook to a real-world production application.